Oksendal 2.10

A stochastic process X_t is called stationary if $\{X_t\}$ has the same distribution as $\{X_{t+h}\}$ for any $h > 0$. Prove that Brownian motion B_t has stationary increments, i.e. that the process $\{B_{t+h} - B_t\}_{h \geq 0}$ has the same distribution for all t.

Solution

To prove this we first need to state the three main properties of brownian motion:

1. Starts at zero: $B_0 = 0$
2. Independent increments: For any $s < t$, the increment $B_t - B_s$ is independent of the process up to time s.
3. Normally distributed increments: For any $s < t$, the increment $(B_t - B_s)$ follows a Normal distribution with mean 0 and variance $t - s$. Mathematically: $(B_t - B_s) \sim N(0, t - s)$.

We want to look at the process $Y_h = B_{t+h} - B_t$ for a fixed t and see if its distribution changes when we change t. We need to show that for any h, the distribution of $B_{t+h} - B_t$ depends only on the time interval h, and not on the starting point t.

Using the third property of brownian motion we can define the distribution of the increment over the interval $[t, t+h]$:

- The length of the interval is $(t + h) - t = h$.
- The increment follows: $(B_{t+h} - B_t) \sim N(0, h)$

Looking at the starting increment from $t = 0$ over the same length of time h we get the following:

- The increment is $B_h - B_0$.
- Since $B_0 = 0$, this is just B_h.
- The distribution: $(B_h - B_0) \sim N(0, h)$

Since $(B_{t+h} - B_t)$ and $(B_h - B_0)$ follow the same Normal distribution $N(0, h)$ regardless of the value of t, the distribution depends only on the gap h, and the starting time t does not affect the distribution. This means that brownian motion has stationary increments.